

INS/RTK GNSS



LiDAR



UWB

Data

# Factor Graph

BP variable nodes

MF variable nodes

Factor nodes in $\mathcal{A}_{\mathrm{BP}}$

Factor nodes in $\mathcal{A} \setminus \mathcal{A}_{\mathrm{BP}}$

$\mathbf{x}_i$

$\mathbf{x}_a$

$\mathbf{m}_{a \to i}$

$\mathbf{n}_{i \to a}$

$f_a$

$\mathbf{m}_{a \to j}$

$\phi_j$

$\phi_a$

Gradient Estimator

Backpropagation

Ground Truths

End-to-End Training

Estimates